AI-Based Collision Avoidance in Crowded Orbits

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1 Background and Summary

The danger of in-space collisions has grown significantly due to the constantly expanding numbers of satellites and orbital junk. Thousands of monitored objects (and hundreds of thousands of smaller fragments) in low Earth orbit (LEO) increase the risk of developing complications such as the Kessler syndrome [1]. Predicting near approaches, estimating the likelihood of a collision, and organizing strategies to avoid collisions are all part of collision avoidance, commonly referred to as conjunction analysis [4]. This procedure has historically depended on orbit propagation models (e.g., SGP4) to predict trajectories, ground-based sensors (radar, telescopes) to catalog objects, and human operators to determine whether to carry out avoidance maneuvers [9].

These manual processes are under stress because to a rise in conjunction alarms per satellite (made worse by megaconstellations), which is driving a move toward more intelligent and automated collision avoidance systems [3]. Recent developments in artificial intelligence (AI) have demonstrated significant promise for improving space situational awareness, especially collision avoidance and orbit prediction [6]. The development of enormous curated databases of conjunction occurrences, the use of machine learning models to forecast collision risk, and the incorporation of artificial intelligence (AI) into satellites for automated maneuver management are some of the major advancements in this domain.

2 Analysis of Research Readings

Recent research in AI-based orbital collision avoidance spans various approaches. Ten notable studies over the past five yeas are summarized below:

- **Machine Learning Competition**: ESA sponsored a challenge using CDM datasets to forecast final collision risk [1]. Ensemble models slightly outperformed a physics-based baseline, but results highlighted the difficulty of the task and importance of feature engineering.
- **Conjunction Screening**: Binary classifiers using neural networks and tree ensembles reduced computational cost and improved conjunction detection versus classical filters [2].

- **Event Pruning**: PCA and clustering filtered out low-risk events, reducing operator burden by up to 70% while preserving safety [3].
- **Decision Support**: Combined ML with Dempster–Shafer theory to classify events and recommend maneuvers under uncertainty [4].
- **Bayesian RNNs**: Early use of RNNs with uncertainty estimation for predicting collision probability from CDM time series [5].
- **Graph Neural Networks**: Proposed a graph-based approach modeling satellite interactions as dynamic links, improving conjunction prediction accuracy [6].
- **Reinforcement Learning**: Developed a DRQN-based system to autonomously trigger fuel-efficient CAMs in simulations [7].
- **Low-Thrust RL**: Demonstrated PPO and DQN agents for maneuvering and reorbiting under low-thrust conditions [8].
- **Multi-Debris Avoidance**: Used genetic algorithms to find optimal maneuvers under simultaneous threats and constraints [6].
- **Literature Review**: Tracked the field's evolution from simple classifiers to ensemble deep models, noting growing interest in explainability and hybrid AI-physics systems [9].

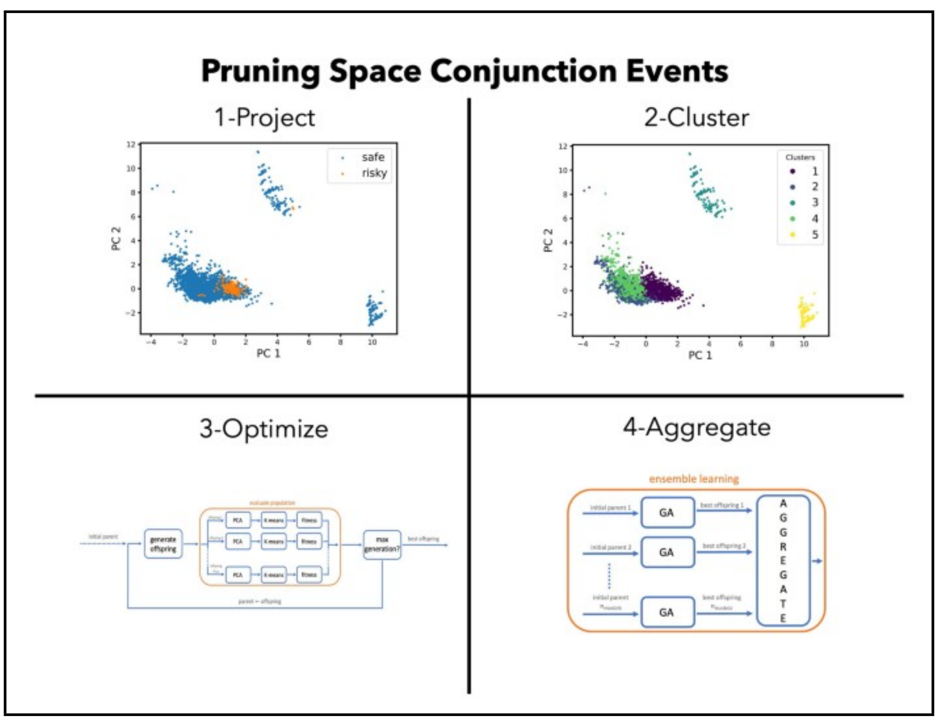

Figure 1: Method for conjunction event pruning (Henry et al., 2023). Classifiers, PCA, and k-means lessen operator strain while preserving security.

The use of ensemble learning for classification, the focus on reducing false alarms, reinforcement learning for autonomy, and Bayesian or evidence-theoretic methods for managing uncertainty are common themes. A paradigm change toward onboard, autonomous conjunction decision systems is suggested by these works.

3 Potential Development Directions

Future studies should concentrate on a few critical topics. On-orbit validation is critical since most modern models rely on synthetic or historical data; active mission demonstrations, like as ESA's CREAM project, are required to demonstrate reliability in real-world circumstances. Furthermore, enhanced orbit prediction is required due to imperfections caused by input ephemerides, demonstrating a clear need for the integration of machine learning with high-fidelity orbit propagation techniques. Multi-agent coordination is still an underexplored subject, highlighting the significance of creating autonomous collision avoidance strategies for large constellations. Furthermore, improving explainability and confidence in AI systems through interpretable models like SHAP values and saliency maps is critical to achieving operator acceptability. Finally, as AI integration progresses, it is critical to address ethical and policy concerns, such as creating fail-safes, overrides, and ensuring conformity with international space traffic norms.

References

[1] C. Uriot et al., "ESA Collision Avoidance Challenge: Predicting Final Collision Risk," *Acta Astronautica*, vol. 198, pp. 121–131, 2022.

[2] D. Stevenson et al., "Machine Learning Methods for Conjunction Screening in Space Traffic Management," *IEEE Trans. Aerospace Electron. Syst.*, vol. 59, no. 1, pp. 78–90, Jan. 2023.

[3] T. Henry et al., "Clustering Approaches for Conjunction Event Pruning," in *Proc. AIAA Scitech*, Jan. 2023.

[4] J. Sánchez and M. Vasile, "Evidence-Theoretic Framework for Collision Avoidance Under Uncertainty," *Journal of Guidance, Control, and Dynamics*, vol. 45, no. 4, pp. 764–777, 2022.

[5] A. Pinto et al., "Bayesian RNNs for Conjunction Event Risk Forecasting," *Advances in Space Research*, vol. 65, no. 3, pp. 902–912, 2020.

[6] Y. Li et al., "Graph Neural Networks for Orbital Conjunction Prediction," arXiv:2303.01045 [cs.LG], 2023.

[7] J. Bourriez et al., "Deep Reinforcement Learning for Autonomous CAM Planning," *Acta Astronautica*, vol. 207, pp. 77–89, 2023.

[8] D. Solomon and T. Păduraru, "Optimized Low-Thrust Collision Avoidance with Deep RL," *IEEE Aerospace Conference*, 2025.

[9] A. Choumos et al., "AI Applications in Space Traffic Management: A Review," *Journal of Space Safety Engineering*, vol. 11, no. 1, pp. 1–15, 2024.

[10] H. D. Curtis, *Orbital Mechanics for Engineering Students*, Butterworth-Heinemann, 2020.